

HEALTHCONNECT CLINIC

Week 7 Project Summary

Project Overview

Week 7 of the HealthConnect Data Analytics project focused on testing, refinement and end-to-end validation of the analytical outputs developed during Weeks 5 and 6. The objective was to assess the accuracy, consistency, usability and decision support value of the existing analysis without rebuilding the existing project

The work followed the progression:

Week 6 Output → Testing → Validation → Refinement → Retesting → Documentation

Objectives

The main objectives for Week 7 were to:

- Review the analytical outputs developed in Week 6.
- Validate important KPI calculations against the underlying HealthConnect dataset.
- Verify that Power BI dashboard values were consistent with the underlying data.
- Test important analytical relationships and segmented findings.
- Review dashboard filters and visual interactions.
- Identify analytical weaknesses, duplication or areas requiring refinement.
- Improve dashboard clarity and decision support value where necessary.
- Retest validated outputs after refinement.

Work Completed

KPI Validation

The five major KPIs carried forward from the previous analysis were independently validated against the underlying HealthConnect appointment data:

The validation confirmed that the calculated results were consistent with the Power BI outputs. Where no discrepancy was identified, the result was recorded as successful validation.

Analytical Validation

The following analytical components were also reviewed and validated:

- Appointment Type
- Booking Lead Group
- Previous No-Show Group
- Reminder Channel
- Selected values from the Booking Lead × Previous No-Show matrix

- Selected values from the Previous No-Show × Reminder Channel matrix

The purpose was to confirm that the patterns presented in Power BI were supported by the underlying records.

Dashboard Testing and Refinement

The Week 6 dashboard was reviewed from both an analytical and usability perspective.

The following components were retained because they provided useful analytical or decision-support information:

- Core KPI cards
- No-Show Rate by Booking Lead Group
- No-Show Rate by Previous No-Show Group
- Booking Lead × Previous No-Show analysis
- Previous No-Show × Reminder Channel analysis
- Reminder Channel analysis
- Appointment Type analysis

The dashboard was also reviewed for unnecessary repetition and visual prioritization. The Appointment Outcome distribution was identified as potentially duplicating information already communicated through the KPI cards. Appointment Day and Distance analyses were reviewed as secondary findings and their prominence was reduced where they provided less decision support value than core analytical relationships.

Slicers were also reviewed to ensure that they served a clear analytical purpose and did not unnecessarily overcrowd the dashboard.

Key Validated Findings

Week 7 testing supported the following findings from the Week 6 analysis:

- Booking Lead Time:** No-show rates differed across booking lead groups, with higher observed rates in the longer booking-lead categories.
- Previous No-Show History:** No-show rates generally increased across higher previous no-show categories. Extreme percentages were considered alongside appointment volume to avoid over interpreting small groups.
- Combined Risk Segmentation:** The Booking Lead × Previous No-Show analysis provided a more detailed view of no-show patterns than analyzing either variable independently.
- Reminder Channel:** Attendance and no-show rates differed across the reminder channel categories. These results were treated as observed associations rather than evidence of causation.

Testing Results

Area	Result	Outcome
KPI validation	Values agreed with Excel	Validated

	validation	
Segmented analysis	Selected calculations agreed with underlying data	Validated
Matrix validation	Selected cells agreed with independent checks	Validated
Filter testing	Relevant filters responded as expected	Validated
Dashboard review	Redundancy/presentation opportunities were reviewed	Refined
Retesting	Important outputs remained consistent	Validated

Decisions and Refinements

Core KPIs and deeper relationship analyses were retained. Redundant or lower-priority visuals were reviewed and reduced where they did not add sufficient decision-support value. The dashboard hierarchy was kept focused on the strongest validated findings.

Cross-Track Testing Contribution

A cross-track testing activity with the Data Science track was planned to assess whether the validated Analytics findings could support future predictive modelling.

Potential inputs identified for further testing included:

- Booking Lead Time
- Previous No-Show History
- Reminder Channel
- Booking Lead Time $\times$ Previous No-Show History

A suitable Data Science collaborator was not available during the Week 7 project period. Therefore, formal cross-track model testing was not completed, and no predictive modelling results were generated.

However, the Analytics findings and proposed modelling inputs were documented as a structured handoff for future integration. This ensures that the analytical work completed during Weeks 6 and 7 can be used as a basis for subsequent cross-track testing.

Challenges and Limitations

The main limitations identified during Week 7 were:

- Some high percentage values may be influenced by small subgroup sizes and therefore require volume context.
- The analysis identifies associations and patterns but does not establish causal relationships.
- Formal Data Science cross-track testing could not be completed because a suitable collaborator was unavailable during the project period.

- Further predictive testing would be required before using the identified variables in a modelling solution.

Conclusion

Week 7 successfully transitioned the HealthConnect Analytics project from advanced analysis to structured testing, refinement and validation. The major KPIs and analytical outputs were checked against the underlying dataset, key relationships were validated, dashboard components were reviewed for clarity and redundancy, and appropriate refinements were identified and tested.

The resulting work provides a more focused and evidence-supported analytical solution while clearly documenting the remaining limitations and cross-track dependencies that can be addressed in subsequent project phases.